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24BCS298

CSE-A1

Implementation of exception handling in Java

Aim:

Write a java program to implement exception handling.

PRE LAB EXERCISE

QUESTIONS

1. What is exception handling in Java?
 - Exception handling in Java is a mechanism to handle runtime errors so that the normal flow of the program is maintained without abrupt termination.
 - It uses keywords like:
`try, catch, finally, throw, throws.`

2. What is the purpose of using try-catch blocks in exception handling?

The purpose of try-catch blocks is to:

- Detect runtime errors
- Handle exceptions gracefully
- Prevent program crash
- Maintain normal program execution
 - try → contains risky code
 - catch → handles the exception if it occurs

IN LAB EXERCISE

Objective

To understand and implement exception handling through try-catch and finally blocks.

Source Code

```
import java.util.InputMismatchException;
import java.util.Scanner;
public class ExceptionHandlingExample {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        try {
            System.out.print("Enter an integer: ");
            int num = scanner.nextInt();
            int result = 10 / num;
            System.out.println("Result: " + result);
        } catch (InputMismatchException e) {
            System.out.println("Invalid input! Please enter an integer.");
        } catch (ArithmeticException e) {
            System.out.println("Cannot divide by zero!");
        } finally {
            scanner.close();
            System.out.println("Program execution completed.");
        }
    }
}
```

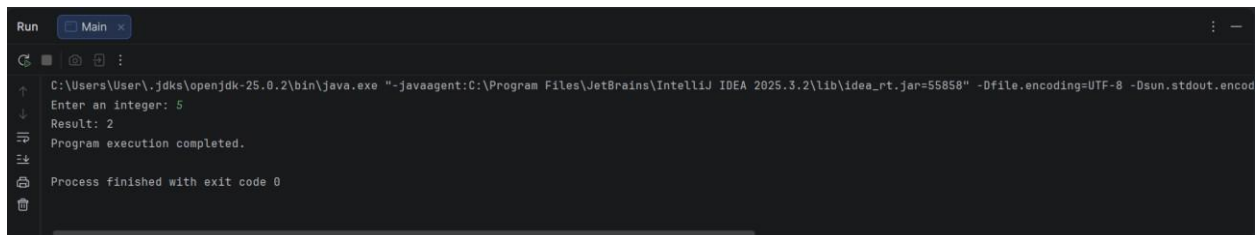
Outputs

Output 1 (Valid Input - Non-zero Integer):

Enter an integer: 5

Result: 2

Program execution completed.

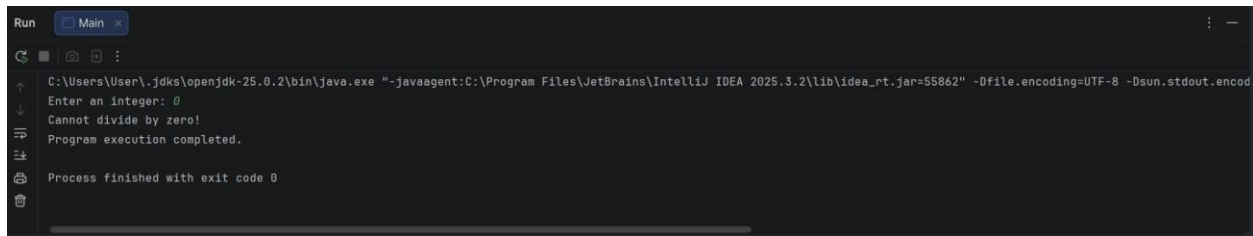


Output 2 (Valid Input - Zero):

Enter an integer: 0

Cannot divide by zero!

Program execution completed.



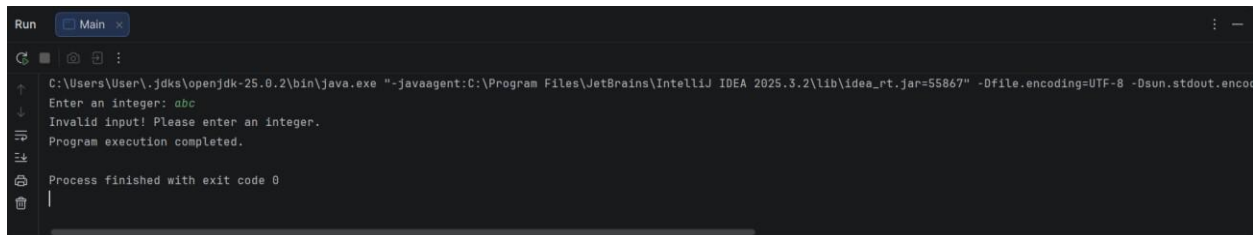
```
Run Main x
C:\Users\User\.jdk\openjdk-25.0.2\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=55862" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8
Enter an integer: 0
Cannot divide by zero!
Program execution completed.
Process finished with exit code 0
```

Output 3 (Invalid Input - Non-integer):

Enter an integer: abc

Invalid input! Please enter an integer.

Program execution completed.



```
Run Main x
C:\Users\User\.jdk\openjdk-25.0.2\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=55867" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8
Enter an integer: abc
Invalid input! Please enter an integer.
Program execution completed.
Process finished with exit code 0
```

POST LAB EXERCISE

1. What is the purpose of the Scanner object in the Java program?

The Scanner object is used to **read input from the user** (keyboard input).

In this program:

- `Scanner scanner = new Scanner(System.in);`
- It reads an integer using `scanner.nextInt();`
So its purpose is to **accept user input during program execution**.

2. What exceptions are expected to be thrown in the code within the try block? Why?

Two exceptions may occur:

1. InputMismatchException

Occurs when:

- The user enters a value that is **not an integer** (e.g., "abc").

Reason:

- nextInt() expects an integer.
 - If the input type is wrong, this exception is thrown.
-

2. ArithmeticException

Occurs when:

- The user enters **0**.

Reason:

- The code performs $10 / \text{num}$.
 - Division by zero is not allowed in Java.
 - So Java throws an ArithmeticException.
-

So the expected exceptions are:

- ✓ InputMismatchException → invalid input
- ✓ ArithmeticException → divide by zero

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		

